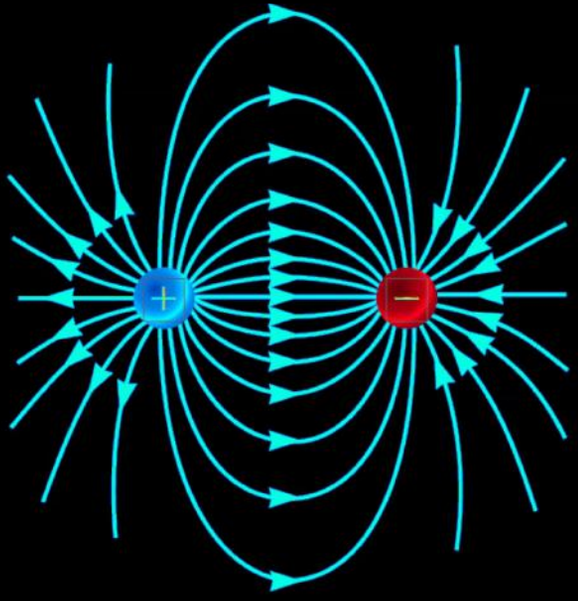
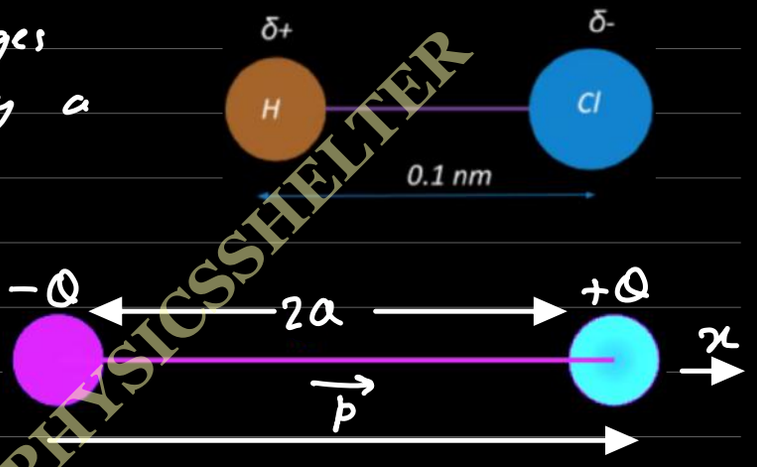
**ELECTRIC DIPOLE:** Two charges of same magnitude but opposite polarity separated by some distance form an electric dipole. (See adjacent figure).

Dipoles are of special interest because dipoles occur naturally in form of molecules with charge separations at atomic levels (e.g. $H^{\delta+}-Cl^{\delta-}$).



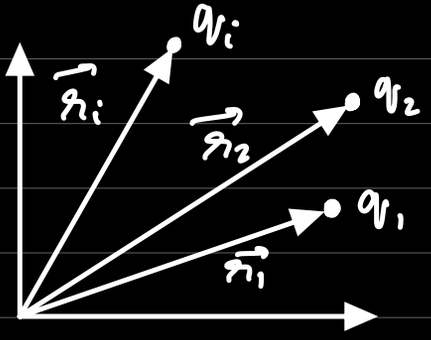
**Dipole Moment Vector:** If two charges $+q$ and $-q$ are separated by a distance $2a$ then we define dipole moment vector as a vector whose magnitude is equal to product of magnitude of one of the charges and the distance between the charges and direction from negative charge to positive charge. For instance for the dipole shown above, $\vec{p} = 2qa\,\hat{i}$




YOUTUBE.COM/@THEPHYSICSSHELTER

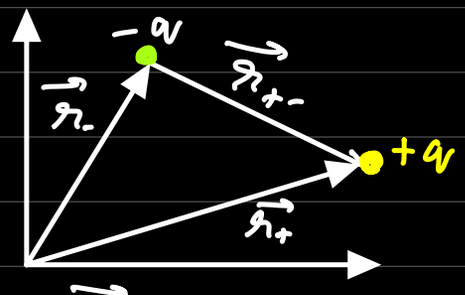
**DIPOLE MOMENT OF A NET NEUTRAL DISTRIBUTION:**

If we have a system of several charges, $q_1, q_2 \ldots q_i \ldots q_n$ located at position vectors, $\vec{r}_1, \vec{r}_2 \ldots \vec{r}_i \ldots \vec{r}_n$ respectively such that $\Sigma q_i = 0$, then dipole moment of distribution is defined as:




$$\boxed{\vec{p} = \Sigma q_i \vec{r}_i}$$

→ One can readily see that this definition nowhere contradicts our first definition, for instance consider a charge pair in adjacent figure, then according to this definition,




$$\vec{p} = q\vec{r}_+ - q\vec{r}_- = q(\vec{r}_+ - \vec{r}_-) = q\vec{r}_{+-}$$

AMBARISH SIR

which is also the dipole moment according to our original definition, thus the new definition is a generalisation of definition of dipole moment to arbitrary net neutral charge distributions.

## Centre Of Charge Method For Finding Dipole Moment :

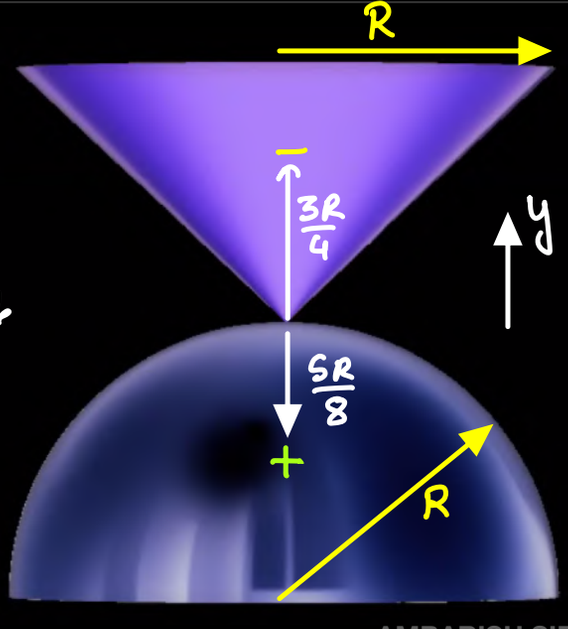
Consider a net neutral distribution of charges such that sum of all positive charges is $+Q$ and that of all negative charges is $-Q$. Let $q_i$ represent $i^{th}$ positive charge and $-q_j$ represent $j^{th}$ negative charge then, according to our definition, dipole moment of the distribution is given as:

$$\vec{p} = \sum q_i \vec{r}_i - \sum q_j \vec{r}_j$$ , this can be also written as:

$$\vec{p} = Q\frac{\sum q_i \vec{r}_i}{Q} - Q\frac{\sum q_j \vec{r}_j}{Q}$$ , Now the quantities $\frac{\sum q_i r_i}{Q}$ and $\frac{\sum q_j r_j}{Q}$ represent centres of positive and negative charges (Similar to centre of mass but with weight factor being charge instead of mass). Thus dipole moment for this distribution can be written as:

$Q(\vec{r}_+ - \vec{r}_-)$ where $\vec{r}_+$ and $\vec{r}_-$ represent position vectors of centres of positive and negative charges.

**Example:** A negatively charged cone of semivertex angle $45°$ is placed over a positively charged hemisphere of radius R as shown in the figure, If magnitude of charge on both is $+Q$. Calculated the dipole moment of the Geometry.



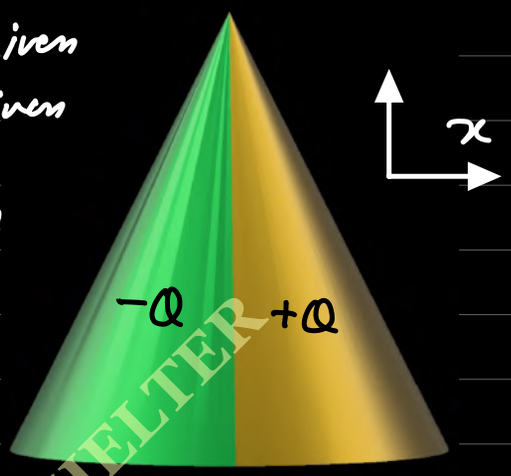

Solution: For hemisphere, $r_+ = \frac{3R}{8}$ from base, i.e $\frac{5R}{8}$ from apex, for cone, $r_- = \frac{3R}{4}$ from base. Concentrating the +ve

YOUTUBE.COM/@THEPHYSICSSHELTER

and negatin charges at these points respectivly, dipole moment:

$$\vec{p} = Q\left(\frac{3R}{4} + \frac{5R}{8}\right)(-\hat{j}) = \frac{11QR}{8}(-\hat{j}) \text{ Ans.}$$

**EXAMPLE:** Half surface of a conical shell is given a negatin charge while the other half is given a positin charge as shown. If the radius of the shell is R. Find the dipole moment of the conical shell if magnitude of individual charges is +Q.

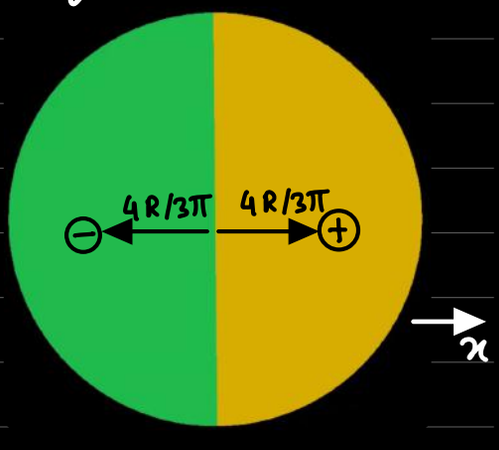



YOUTUBE.COM/@THEPHYSICSSHELTER

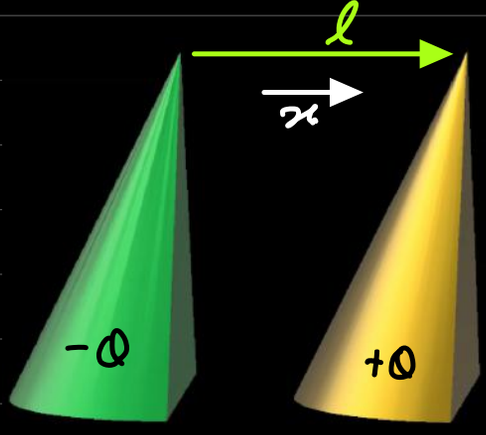
Solution: By symmetry it can be readily seen that net dipole moment of cone is in +x direction. Now if we deform the cone in such a way that x coordinate of any charge does not change then x component of dipole will not change. Now suppose we squeeze the cone by moving the particles ∥ to y axis we have two uniformly charged discs. (You can also think that we have just translated pairwise dipoles parallel to themselves, thus dipole moment is still conserved).

Now concentrating charges at centres of charge, dipole moment is simply:

$$\vec{p} = \frac{8R}{3\pi} Q\,\hat{i}$$

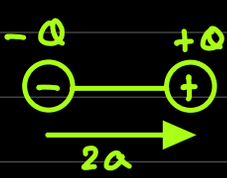



**EXAMPLE:** Two identically shaped half solid cones are placed in identical orientations as shown. If both have a uniform charge distributions, find the magnitude of dipole moment of the system.




Here, every +q, −q pair can be imagined to be separated by ℓ ∴ $\vec{p} = Q\ell\,\hat{i}$ Ans.

**POINT DIPOLE:** Point dipole is an idealized dipole whose size tends to zero and the individual charges tend to $\infty$ such that the product $2Qa$ is still finite. In practice point dipoles do not exist however at relatively larger distances, all dipoles can be approximated as point dipoles as we shall see in later sections.




**FIELD DUE TO DIPOLE IN AXIAL POSITION:**

A point on line joining the charges of the dipole is said to be lying in axial position. Consider field at point A due to the charges of dipole, Due to +Ve charge,

$\vec{E_1} = \frac{kq}{(r-a)^2}\hat{p}$ Similarly due to negatin charge,

$\vec{E_2} = \frac{kq}{(r+a)^2}(-\hat{p})$, Thus net field:

$$\vec{E} = \vec{E_1} + \vec{E_2} = kq\hat{p}\left[\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2}\right] = \frac{4kqar}{(r^2-a^2)^2}\hat{p}$$

Also $\vec{p} = 2qa\hat{p}$ thus, Field at axial position:

$$\boxed{\vec{E_a} = \frac{2k\vec{p}\,r}{(r^2-a^2)^2}}$$

, Now if $r >> a$ (i.e point dipole) we can ignore $a$ in denominator

Thus for point dipoles: $\boxed{\vec{E_a} = \frac{2k\vec{p}}{r^3}}$

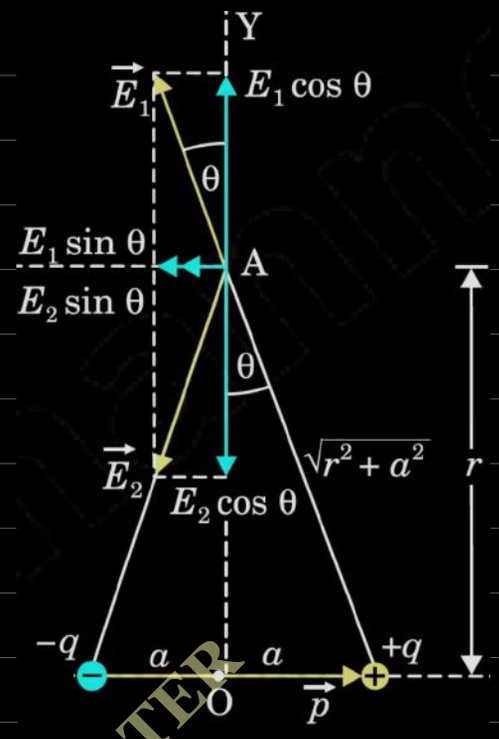
→ Note that field of dipole diminishes with $1/r^3$ and not $1/r^2$

→ Direction of field on the axis is same as that of the dipole moment.

→ Any net neutral distribution when viewed from a very large distance will behave like point dipole, thus the formula for point dipoles can be used to estimate fields due to extended neutral distributions at large distances.

YOUTUBE.COM/@THEPHYSICSSHELTER

## FIELD DUE TO DIPOLE AT ⊥ BISECTOR (EQUATOR) :




Let A be a point on ⊥ bisector of a dipole at a distance $r$ from the mid point of the dipole. By Symmetry it can be seen that components of fields ∥ to dipole add up while those ⊥ to dipole cancel out.

As clear from figure,

$$\vec{E}_{1\parallel} = \vec{E}_{2\parallel} = \frac{kq}{(r^2+a^2)} \sin\theta\,(-\hat{p})$$

Also $\sin\theta = a/\sqrt{r^2+a^2}$, Thus net field on equator:

$$\vec{E} = \vec{E}_{1\parallel} + \vec{E}_{2\parallel} = \frac{2kqa}{(r^2+a^2)^{3/2}}(-\hat{p}) \;;\; \text{Also } \vec{p} = 2qa\hat{p}$$

Thus:

$$\boxed{\vec{E}_p = \frac{-k\vec{p}}{(r^2+a^2)^{3/2}}}$$

Now if $r >> a$, we can ignore $a$,

Thus for point dipole, at equatorial position (⊥ bisector):

$$\boxed{\vec{E}_p = \frac{-k\vec{p}}{r^3}}$$

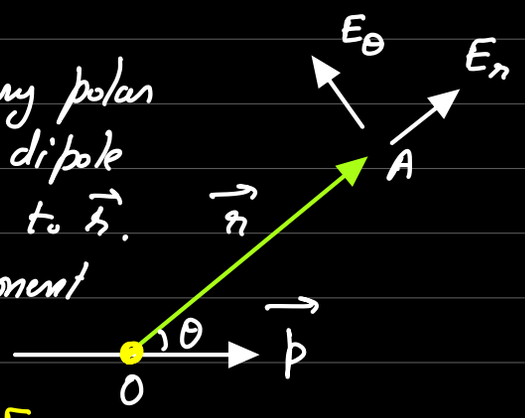
→ For same distance, field at equator has half the magnitude of field as that on axial position at same distance.

→ Field at equator is anti-parallel to the dipole moment vector (whereas at axial position it was parallel).

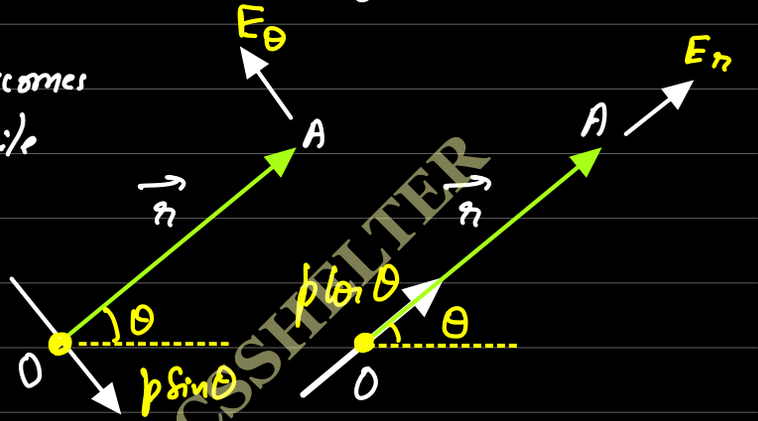
YOUTUBE.COM/@THEPHYSICSSHELTER

# FIELD DUE TO POINT DIPOLE IN ARBITRARY ORIENTATION (Vector Resolving).

Field due to a point dipole at an arbitrary polar coordinate can be found by resolving its dipole moment vector in to components $\parallel$ and $\perp$ to $\vec{r}$. The $\parallel$ component gives us the radial component and $\perp$ component of dipole gives the tangential component.




As clear from adjacent figures, A becomes axial point for component $p\cos\theta$ while equatorial point for $p\sin\theta$, thus the net electric field components at A are given as:




$$\boxed{\vec{E}_r = \frac{2kp\cos\theta}{r^3}\hat{r}}$$ and

$$\boxed{\vec{E}_\theta = \frac{kp\sin\theta}{r^3}\hat{\theta}}$$

$$E = \sqrt{E_r^2 + E_\theta^2} = \frac{kp}{r^3}\sqrt{1+3\cos^2\theta}$$

$$\boxed{E = \frac{kp}{r^3}\sqrt{1+3\cos^2\theta}}$$

A rigorous proof of why the vector resolution words will be given in a later section. (Using potential arguments)

**FIELD DUE TO A DIPOLE IN 3D:** In the previous section we calculated electric field due to dipole in a particular plane, however due to cylindrical symmetry, the same kind of field will exist on any circle coaxial with dipole axis. We shall try to extend the formulae derived in previous section to 3D using the vector ideas. (Trying to generate sine using cross product and cosine using dot product). From previous section, we have:

$$\vec{E} = \frac{2kp\cos\theta}{r^3}\hat{r} + \frac{kp\sin\theta}{r^3}\hat{\theta} \quad (1)$$

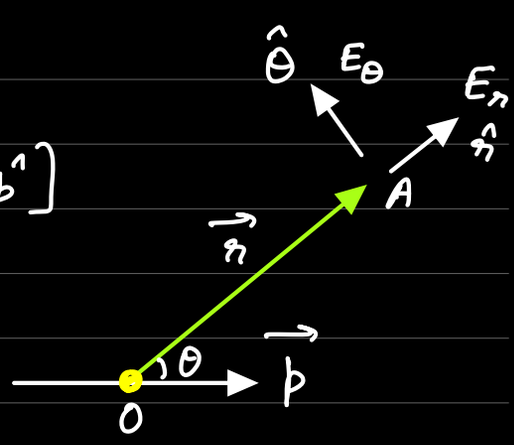
YOUTUBE.COM/@THEPHYSICSSHELTER

Now realise that, $\cos\theta = \hat{p}\cdot\hat{n}$ ② and

$\sin\theta\,\hat{\theta} = (\hat{p}\times\hat{n})\times\hat{n} = -\hat{n}\times(\hat{p}\times\hat{n})$

$\Rightarrow \sin\theta\,\hat{\theta} = -[\hat{p}(\hat{n}\cdot\hat{n}) - \hat{n}(\hat{n}\cdot\hat{p})] = [(\hat{n}\cdot\hat{p})\hat{n} - \hat{p}]$

$\Rightarrow \sin\theta\,\hat{\theta} = [(\hat{n}\cdot\hat{p})\hat{n} - \hat{p}]$ ③




Now using ② & ③ in ①:

$$\vec{E} = \frac{k}{n^3}\left[2(\hat{p}\cdot\hat{n})\hat{n} + (\hat{n}\cdot\hat{p})\hat{n} - \hat{p}\right]$$

$$\Rightarrow \vec{E} = \frac{k}{n^3}\left[3(\hat{p}\cdot\hat{n})\hat{n} - \hat{p}\right] \Rightarrow \boxed{\vec{E} = \frac{k}{n^3}\left[3(\vec{p}\cdot\hat{n})\hat{n} - \vec{p}\right]} \quad ④$$

**VALUE OF $\theta$ FOR $\vec{E}\perp\vec{p}$** : For $\vec{E}\perp\vec{p}$, $\vec{E}\cdot\vec{p} = 0$, using this in equation ④ above.

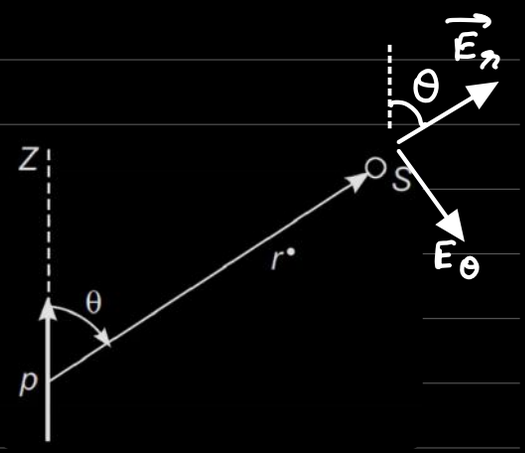
$$3(\vec{p}\cdot\hat{n})(\hat{n}\cdot\vec{p}) - p^2 = 0 \Rightarrow 3p^2\cos^2\theta - p^2 = 0$$

$\Rightarrow \cos\theta = \pm\frac{1}{\sqrt{3}}$ or $\tan\theta = \pm\sqrt{2}$. Thus for perpendicular field:

$$\boxed{\theta = \cos^{-1}(\pm 1/\sqrt{3}) = \tan^{-1}(\pm\sqrt{2})}$$

**EXAMPLE (IRODOV 3.40)**

A dipole with electric moment $\vec{p}$ oriented in +ve z direction is located at origin of coordinates, Find the projections $E_z$ and $E_\perp$ of electric field strength vector. At which point is $\vec{E}\perp\vec{p}$?




Solution: Refer to the figure above, $E_{11} = E_n\cos\theta - E_\theta\sin\theta$

$$\Rightarrow E_{11} = \frac{2kp\cos^2\theta}{n^3} - \frac{kp\sin^2\theta}{n^3} = \frac{kp}{n^3}\left[3\cos^2\theta - 1\right] \text{ abs.}$$

$$E_\perp = E_n\sin\theta + E_\theta\cos\theta = \frac{2kp\sin\theta\cos\theta}{n^3} + \frac{kp\sin\theta\cos\theta}{n^3}.$$

$$\Rightarrow \boxed{E_\perp = \frac{3kp\sin\theta\cos\theta}{n^3}} \text{ abs.} \qquad \boxed{E_{11} = \frac{kp}{n^3}\left[3\cos^2\theta - 1\right]}$$

→ Note that $E_\perp$ has maximum magnitude for $\theta = \pm 45^\circ, \pm 135^\circ$.

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

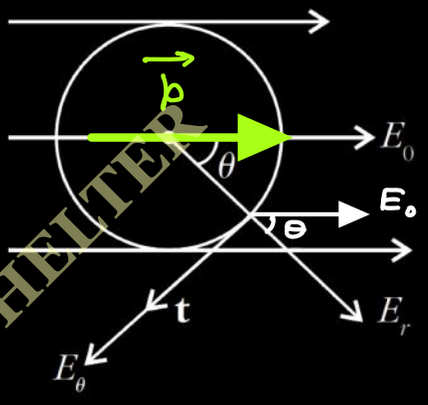
→ $E_\parallel$ vanishes at $\cos^{-1}(\pm 1/\sqrt{3})$ or $\tan^{-1}(\pm\sqrt{2})$ as seen in the previous section, at this position, $E_\perp = \sqrt{2}\,kp/r^3$.

## Example: IrodOv 3.41

A point electric dipole with a moment $\vec{\mathbf{p}}$ is placed in the external uniform electric field whose strength equals $\vec{\mathbf{E}}_0$, with $\vec{\mathbf{p}} \uparrow\uparrow \vec{\mathbf{E}}_0$. In this case one of the equipotential surfaces enclosing the dipole forms a sphere. Find the radius of this sphere.

Solution: Field lines are always perpendicular to an equipotential surface, thus at the equipotential surface, the tangential field due to dipole must be cancelled by that of uniform external electric field, thus:

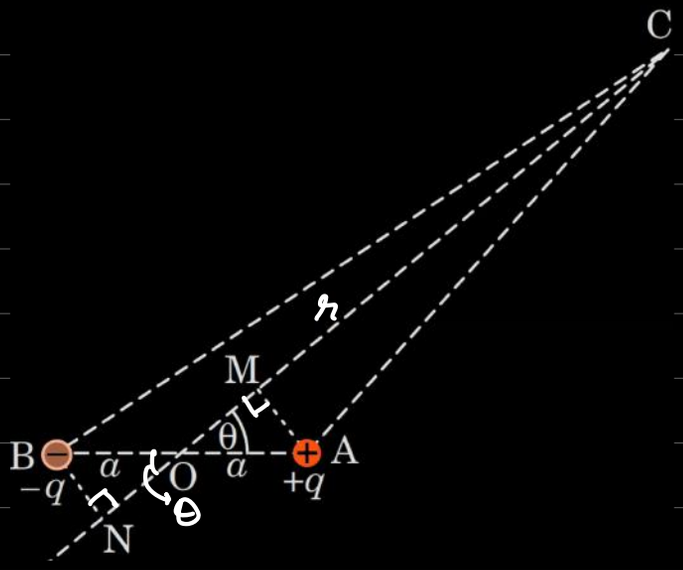
$$\frac{kp\sin\theta}{r^3} = E_0\sin\theta \Rightarrow r = \left(\frac{kp}{E_0}\right)^{1/3} \text{ Ans.}$$




(This problem can also be done using concept of potential due to a dipole, however we shall use that approach after learning the potential due to dipole).

## Potential Due To A Point Dipole At Arbitrary Point:

Consider a short dipole as shown in the figure. We wish to find potential due to this at a distant point C. Let distance of C from mid point of dipole be $OC = r$. Now to estimate AC we place compass at C and make an arc with radius AC to cut OC at M thus $AC = CM = r - a\cos\theta$, Similarly, $BC = r + a\cos\theta$. Thus potential at C:




$$V = \frac{kq}{r - a\cos\theta} - \frac{kq}{r + a\cos\theta} = \frac{2kqa\cos\theta}{r^2 - a^2\cos^2\theta} = \frac{kp\cos\theta}{r^2 - a^2\cos^2\theta}$$

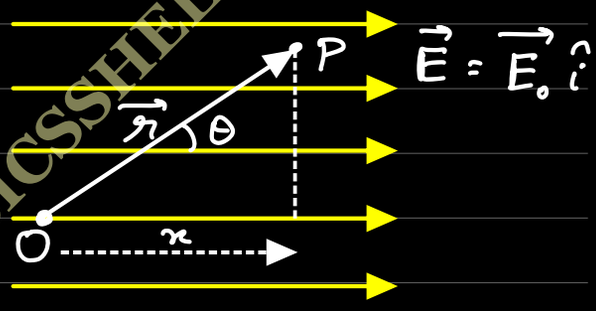
For point dipole, $a = 0 \Rightarrow$ $\boxed{V = \frac{kp\cos\theta}{r^2}}$, also: $\boxed{V = \frac{k\vec{p}\cdot\vec{r}}{r^3}}$

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

Now we can find radial and axial fields due to this dipole using the relations, $E_r = -\frac{\partial V}{\partial r}$ and $E_\theta = -\frac{1}{r}\frac{\partial V}{\partial \theta}$, thus we obtain: $E_r = \frac{2kp\cos\theta}{r^3}$ and $E_\theta = \frac{kp\sin\theta}{r^3}$, these are the same results that we earlier obtained using vector resolution of point dipole. Thus vector resolution is perfectly legitimate method for finding fields due to point dipoles.

## Potential In A Uniform Electric Field :

Consider a point P in a uniform field whose position vector is $\vec{r}$ wrt some point O.



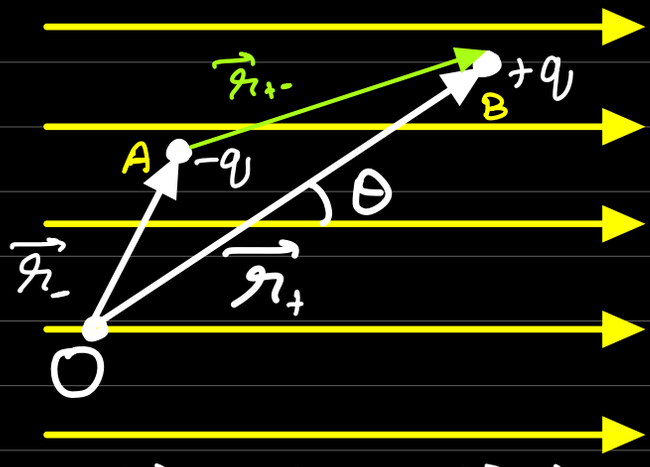

Now $E_x = -\frac{\partial V}{\partial x} = E_0 \Rightarrow V = -E_0 x + C$

$\Rightarrow V = -E_0 r\cos\theta + C = -\vec{E_0}\cdot\vec{r} + C$,

Where constant C depends on the choice of zero potential. (If O is taken as reference then $C = 0$). However we can always work with help C as arbitrary constant as well.

## Potential Energy Of A Dipole In Uniform Field:

Recall that potential energy of any charged particle is given as $U = qV$. Now consider a dipole in a uniform field, as shown in the figure. Now:




$$U_+ = q(-\vec{E}\cdot\vec{r}_+ + C)$$

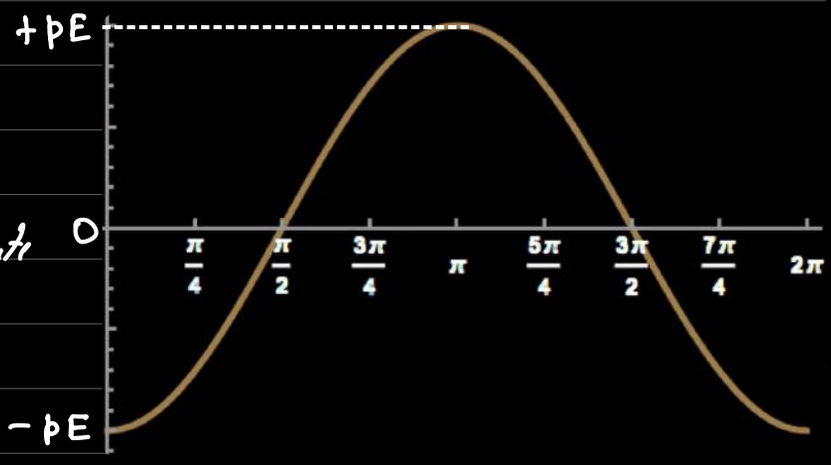
$$U_- = -q(-\vec{E}\cdot\vec{r}_- + C)$$

$$\Rightarrow U = U_+ + U_- = q\left[-\vec{E}\cdot(\vec{r}_+ - \vec{r}_-)\right] = -\vec{E}\cdot q\vec{r}_{+-} = -\vec{p}\cdot\vec{E}$$

Thus $\boxed{U = -\vec{p}\cdot\vec{E}}$ (Note that this does not include interaction energy of point charges).

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

→ Min PE occurs for $\theta = 0°$ i.e $\vec{E} \parallel \vec{p}$ and max pE occurs for $\theta = 180°$ is $\vec{E} \uparrow\downarrow \vec{p}$. When a dipole is released from arbitrary position, it will generally oscillate about stable equilibrium position.



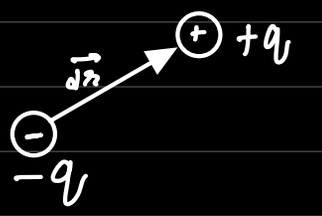

→ Recall from our discussion in SHM in conservative force fields that:

$T = 2\pi \sqrt{\frac{m}{U''(x_0)}}$ for linear oscillations similarly for angular oscillations,

$T = 2\pi \sqrt{\frac{I}{U''(\theta_0)}}$ for angular oscillations, where $\theta_0$ is stable equilibrium orientation angle, I is moment of inertia about axis of Rotation. Thus for dipole, $\boxed{T = 2\pi \sqrt{I/pE_0}}$

## Potential Energy Of A Point Dipole In General Electric Field :

Consider a dipole with position vector of the charge as $d\vec{r}$ wrt negative charge.




$\Rightarrow U_+ = q V_+ \quad U_- = -q V_-$

$\Rightarrow U = U_+ + U_- = q(V_+ - V_-) = q \frac{(V_+ - V_-)}{dr} dr$

Now recall that $-\frac{\partial V}{\partial r}$ represents field Electric field component in $\hat{r}$ direction

Thus $-\left(\frac{V_+ - V_-}{dr}\right)$ represents component of Electric field along $d\vec{r}$ direction.

i.e. $(V_+ - V_-)/dr = -\vec{E} \cdot d\hat{r} \Rightarrow U = -q \vec{E} \cdot d\hat{r}\, dr$

$\Rightarrow U = -\vec{E} \cdot q d\vec{r} = -\vec{p} \cdot \vec{E}$. Thus potential energy of a dipole in arbitrary field is also given as:

$$\boxed{U = -\vec{p} \cdot \vec{E}}$$

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

## Force On A Dipole In Arbitrary Field:

Recall from our discussion in work power and energy that $\vec{F} = -\nabla U$ i.e: $F_x = -\frac{\partial U}{\partial x}$, $F_y = -\frac{\partial U}{\partial y}$ and $F_z = -\partial U/\partial z$. Now let there be a point dipole in an electric field given by $p_x\hat{i} + p_y\hat{j} + p_z\hat{k}$ and let the electric field be $E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$. Then:

$$U = -\vec{p}\cdot\vec{E} = -p_x E_x - p_y E_y - p_z E_z.$$

Note that for a dipole in a particular orientation $p_x$, $p_y$ and $p_z$ are constants. Thus $F_x = p_x\frac{\partial E_x}{\partial x} + p_y\frac{\partial E_y}{\partial x} + p_z\frac{\partial E_z}{\partial x}$ (1) $\left(\because -\frac{\partial U}{\partial x}\right)$

Also recall that for a conservative field, $\frac{\partial E_y}{\partial x} = \frac{\partial E_x}{\partial y}$ and also,

$\frac{\partial E_z}{\partial x} = \frac{\partial E_x}{\partial z} \Rightarrow \vec{F_x} = p_x\frac{\partial E_x}{\partial x} + p_y\frac{\partial E_x}{\partial y} + p_z\frac{\partial E_x}{\partial z}\,\hat{i}$ (2)

Similarly, $\vec{F_y} = p_x\frac{\partial E_y}{\partial x} + p_y\frac{\partial E_y}{\partial y} + p_z\frac{\partial E_y}{\partial z}\,\hat{j}$ and

$$\vec{F_z} = p_x\frac{\partial E_z}{\partial x} + p_y\frac{\partial E_z}{\partial y} + p_z\frac{\partial E_z}{\partial z}\,\hat{k}$$

Thus net force can be compactly written as:

$$\boxed{\vec{F} = (\vec{p}\cdot\nabla)\vec{E}} \quad (3)$$

Here $\vec{p}\cdot\nabla$ is the operator $(p_x\hat{i} + p_y\hat{j} + p_z\hat{k})\cdot[\hat{i}\,\partial/\partial x + \hat{j}\,\partial/\partial y + \hat{k}\,\partial/\partial z]$

i.e. $[p_x\,\partial/\partial x + p_y\,\partial/\partial y + p_z\,\partial/\partial z]$, Thus:

$$\vec{F} = [p_x\,\partial/\partial x + p_y\,\partial/\partial y + p_z\,\partial/\partial z]\,[E_x\hat{i} + E_y\hat{j} + E_z\hat{k}]$$

Collecting $\hat{i}$ coefficients, $F_x = p_x\frac{\partial E_x}{\partial x} + p_y\frac{\partial E_x}{\partial y} + p_z\frac{\partial E_x}{\partial z}$ this is same as equation (2)

Note that (3) can also be written as: $\boxed{\vec{F_x} = (\vec{p}\cdot\vec{\nabla})E_x}$ etc

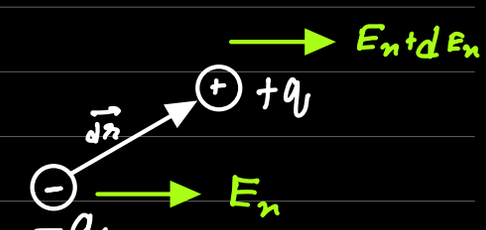
YOUTUBE.COM/@THEPHYSICSSHELTER

## Alternative Derivation for Force on a dipole:

Consider the dipole shown in the figure,

$F_x = q(E_x + dE_x) - q(E_x)$

$F_x = q\,dE_x$. Now by Taylor approximation,




$$dE_x = \frac{\partial E_x}{\partial x}dx + \frac{\partial E_x}{\partial y}dy + \frac{\partial E_x}{\partial z}dz \quad (\because \text{Taylor}).$$

Also $\vec{p} = q[dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}] \Rightarrow dx = \frac{p_x}{q}$, $dy = \frac{p_y}{q}$, $dz = \frac{p_z}{q}$

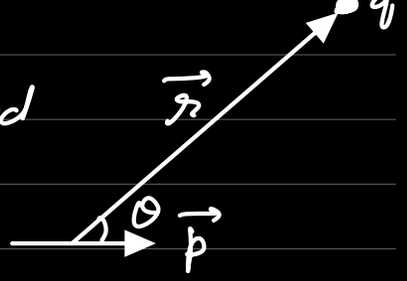
$$\Rightarrow F_x = q\left[\frac{\partial E_x}{\partial x}\cdot\frac{p_x}{q} + \frac{\partial E_x}{\partial y}\cdot\frac{p_y}{q} + \frac{\partial E_x}{\partial z}\cdot\frac{p_z}{q}\right]$$

$$\Rightarrow F_x = p_x\frac{\partial E_x}{\partial x} + p_y\frac{\partial E_x}{\partial y} + p_z\frac{\partial E_x}{\partial z}$$, this is same as equation (1) above.

Note: $(\vec{p}\cdot\vec{\nabla})\vec{E} = -\nabla(-\vec{p}\cdot\vec{E}) = \nabla(\vec{p}\cdot\vec{E})$ if $\vec{p}$ is a constant vector and $\vec{E}$ is a conservative field.

## Force Between A Point Charge And Dipole:

We could find force between a point charge and dipole using several methods (such as $\vec{F} = (\vec{\nabla}\cdot\vec{p})\vec{E}$ or $\vec{F} = -\nabla U$), however since we already know the field due to dipole, most efficient method would be simply $\vec{F} = q\vec{E}$, where, $\vec{E} = \frac{2kp\cos\theta}{r^3}\hat{r} + \frac{kp\sin\theta}{r^3}\hat{\theta}$



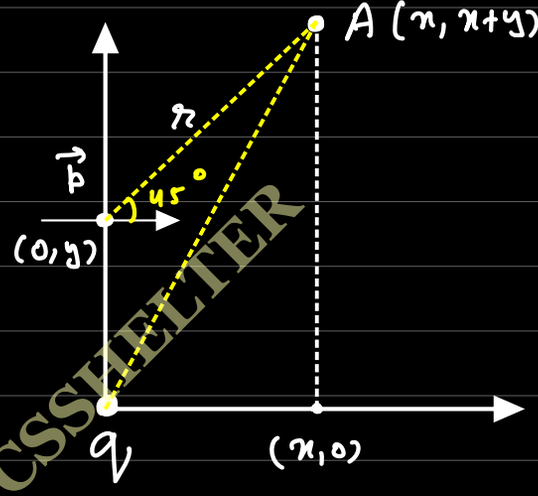

This will give us force on point charge and force on dipole will be just negative of this. For 3D fields due to dipole recall that $\vec{E} = \frac{k}{r^3}[3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}]$.

YOUTUBE.COM/@THEPHYSICSSHELTER

## EXAMPLE: PATHFINDER MCQ-7

If a dipole of dipole moment $p\hat{i}$ is placed at point $(0, y)$ and a point charge at the origin of a coordinate system, net electric field at point $(x, x + y)$ vanishes. If $x$ and $y$ both are positive, the coordinate $y$ is equal to

(a) $x$ (b) $2x$
(c) $2.5x$ (d) $3x$




For cancelling fields at point A, directions of fields due to $q$ and $\vec{p}$ must be opposite. (i.e ratio of $y$ and $x$ components must be same).

Now field due to $q$ is along (against) $\vec{E}_q = C(x\hat{i} + (x+y)\hat{j})$.

Now recall from a previous example,

$$E_{\parallel} = \frac{kp}{r^3}(3\cos^2\theta - 1) \text{ and } E_{\perp} = 3\frac{kp}{r^3}\sin\theta\cos\theta$$

Thus due to dipole, $E_x = \frac{kp}{r^3}[3\cos^2 45° - 1] = \frac{1}{2}\frac{kp}{r^3}$,

$$E_y = \frac{3kp}{r^3}[\sin 45° \cos 45°] = \frac{3}{2}\frac{kp}{r^3} \Rightarrow \frac{E_y}{E_x} = 3,$$

For directions of field due to point charge and dipole to be antiparallel, this must also be the ratio of $E_y/E_x$ due to point charge thus: $3 = (x+y)/x \Rightarrow y = 2x$ Ans.

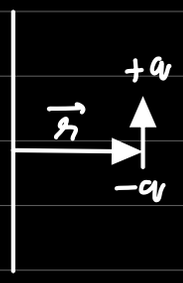
## EXAMPLE: IRODOV 3.46 :

A dipole with an electric moment **p** is located at a distance $r$ from a long thread charged uniformly with a linear density $\lambda$. Find the force **F** acting on the dipole if the vector **p** is oriented

(a) along the thread;
(b) along the radius vector **r**;
(c) at right angles to the thread and the radius vector **r**.

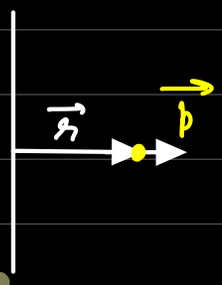
YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: (a) Here since both charges are equidistant from the line charge on a line ⊥ to the line charge, forces on the dipole charges are equal and opposite ∴ $\vec{F}=0$.



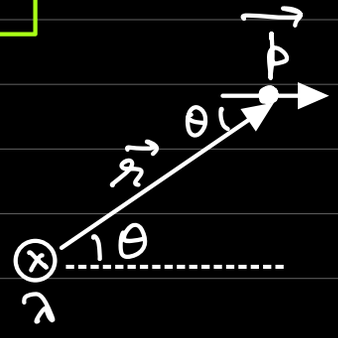

(b) Here as shown in adjacent figure, the negative charge experiences a stronger pull than the repulsion faced by +Ve charge. Now, for the dipole oriented along the field, $U=-\vec{p}\cdot\vec{E}$

$$\Rightarrow U=-(p\hat{r})\cdot\left(\frac{\lambda}{2\pi\epsilon_0 r}\hat{r}\right)=\frac{-\lambda p}{2\pi\epsilon_0 r}$$




Now $F_r=-\frac{\partial U}{\partial r}=\frac{\lambda p}{2\pi\epsilon_0}[-1\,\hat{r}^2]\Rightarrow \boxed{\vec{F}=\frac{-\lambda\vec{p}}{2\pi\epsilon_0 r^2}}$

(c) Here dipole is in a plane ⊥ to the line charge, for better visualisation, we drawn a view where line charge appears to be a point, and we consider a dipole making an angle θ with electric field at position vector $\vec{r}$ (We shall find force in this orientation and then put θ = 90°.




Now $U=-\vec{p}\cdot\vec{E}=-[p\cos\theta\hat{r}-p\sin\theta\hat{\theta}]\cdot\left[\frac{\lambda}{2\pi\epsilon_0 r}\hat{r}\right]=\frac{-\lambda p\cos\theta}{2\pi\epsilon_0 r}$

Now $F_r=-\frac{\partial U}{\partial r}\hat{r}=\frac{\lambda p\cos\theta}{2\pi\epsilon_0}[-1\,\hat{r}^2]=\frac{-\lambda p\cos\theta}{2\pi\epsilon_0 r^2}$

$$F_\theta=\frac{-1}{r}\frac{\partial U}{\partial\theta}=\frac{\lambda p}{2\pi\epsilon_0 r^2}[-\sin\theta]=\frac{-\lambda p\sin\theta}{2\pi\epsilon_0 r^2}$$

Now for $\theta=90°$, $F_r=0$, $\vec{F}_\theta=\frac{-\lambda p}{2\pi\epsilon_0 r^2}\hat{\theta}$ Ans.

For magnitude of force on a skew ⊥ dipole ($\vec{p}\perp\vec{r}$):

$$\boxed{F=F_\theta=\frac{\lambda p}{2\pi\epsilon_0 r^2}}$$

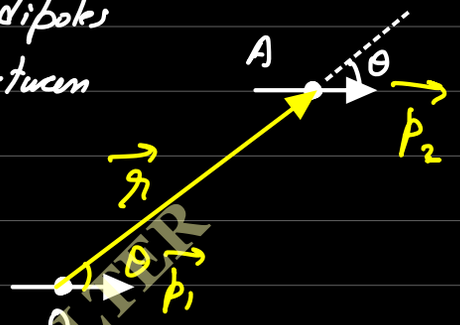
→ Note that tangential and radial force expressions are as if there were two dipoles $p\cos\theta$ & $p\sin\theta$ (components), one of them is experiencing only radial force and other one only tangential force.

AMBARISH SIR
YOUTUBE.COM/@THEPHYSICSSHELTER
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

→ Magnitude of answers both (b) & (c) part is $\frac{2p}{2\pi\epsilon_0 r^3}$.

→ For a dipole in arbitrary orientation, its components can be taken and then force on each component can be calculated by comparing with cases (a), (b) and (c) as required.

**FORCE BETWEEN ⊥ dipoles:** Consider two parallel dipoles as shown in the figure such that $\theta$ is angle between $\vec{r}$ and $\vec{p}_2$. Now due to dipole $\vec{p}_1$, field at A is given as $\vec{E} = \frac{2kp_1\cos\theta}{r^3}\hat{r} + \frac{kp\sin\theta}{r^3}\hat{\theta}$




also $\vec{p}_2 = p_2\cos\theta\hat{r} - p_2\sin\theta\hat{\theta}$, Now:

$$U = -\vec{p}_2\cdot\vec{E} = \frac{-kp_1p_2}{r^3}\left[2\cos^2\theta - \sin^2\theta\right]$$

Now $F_r = -\frac{\partial U}{\partial r} = kp_1p_2(2\cos^2\theta - \sin^2\theta)\left[-3r^{-4}\right] = \frac{-3kp_1p_2}{r^4}\left[3\cos^2\theta - 1\right]$

$$F_\theta = \frac{-1}{r}\frac{\partial U}{\partial\theta} = \frac{kp_1p_2}{r^4}\left[6\cos\theta(-\sin\theta)\right] = -\frac{6kp_1p_2}{r^4}\sin\theta\cos\theta$$

**FORCE BETWEEN COAXIAL DIPOLES:**

Force in this configuration corresponds to $\theta = 0°$ in the above derivation for ⊥ dipoles, accordingly, $F_\theta = 0$

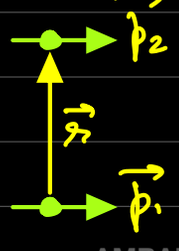



$\therefore \vec{F} = \vec{F}_r = -\frac{6kp_1p_2}{r^4}\hat{r}$

$$\boxed{F = \frac{6kp_1p_2}{r^4}}$$ ← Memorize magnitude, put direction by common sense. (Like coaxial dipoles attract)

**FORCE BETWEEN ⊥ DIPOLES ($\vec{p} \perp \vec{r}$):**

Force in this configuration can be easily found by setting $\theta = 90°$ in the general result for force between parallel dipoles, thus, $F_\theta = 0$, $\boxed{F = \frac{3kp_1p_2}{r^4}} = F_r$



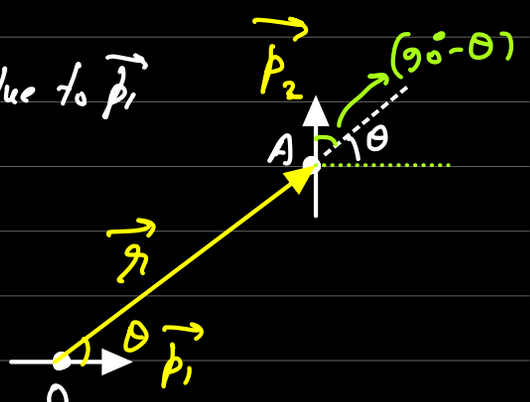

AMBARISH SIR

YOUTUBE.COM/@THEPHYSICSSHELTER

→ Like dipoles repel each other in this orientation.

**FORCE BETWEEN ⊥ DIPOLES:** Electric field at A due to $\vec{p}_1$

$$\vec{E} = \frac{2kp_1\cos\theta}{r^3}\hat{r} + \frac{kp_1\sin\theta}{r^3}\hat{\theta}$$




$$\vec{p}_2 = p_2\sin\theta\hat{r} + p_2\cos\theta\hat{\theta}$$

$$\Rightarrow U = -\vec{p}_2\cdot\vec{E} = \frac{-kp_1p_2}{r^3}[3\sin\theta\cos\theta]$$

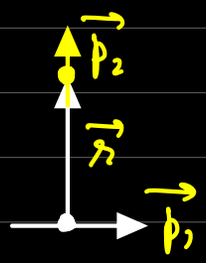
$$\Rightarrow F_r = -\frac{\partial U}{\partial r} = 3kp_1p_2\sin\theta\cos\theta\,[-3r^{-4}] = -\frac{9kp_1p_2}{r^4}\sin\theta\cos\theta$$

$$\vec{F}_\theta = -\frac{1}{r}\frac{\partial U}{\partial\theta}\hat{\theta} = \frac{3kp_1p_2}{r^4}[\cos 2\theta]\hat{\theta} \quad (\because 3\sin\theta\cos\theta = 3/2\sin 2\theta).$$

**FORCE BETWEEN T SHAPE ⊥ DIPOLES ($\vec{r}\perp$ one of the $\vec{p}$ & $\parallel$ to other).**

Force in this case can be easily found by putting $\theta = 90°$ in previous derivation, Accordingly:

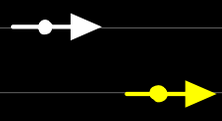
$$\vec{F}_r = 0, \quad \therefore F = F_\theta = \frac{3kp_1p_2(-\hat{\theta})}{r^4}$$




Thus $\boxed{F = \frac{3kp_1p_2}{r^4}}$ → Force is ⊥ to $r$.

→ Direction of force can be found by looking whether ⊕ charge of $p_2$ is in stronger field or ⊖ charge in stronger field.
For instance for the figure shown above, +Ve charge experiences smaller force, thus net force on $p_2$ is to the right.

**SUMMARY OF FORCE ON DIPOLES:**



$F = 6kp_1p_2/r^4$



$F = 3kp_1p_2/r^4$



$F = 3kp_1p_2/r^4$

YOUTUBE.COM/@THEPHYSICSSHELTER

→ For arbitrary orientations of two dipoles, both dipoles should be resolved ∥ and ⊥ to the $\vec{R}$. Now net force must be found by superposition of following four pairwise interactions with proper attention to sign of interactions, namely,

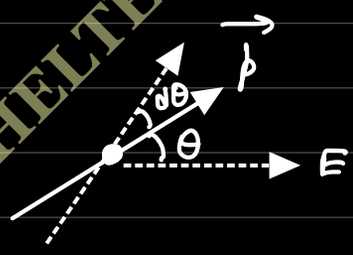
$(p_{1\parallel}, p_{2\parallel})$; $(p_{1\parallel}, p_{2\perp})$, $(p_{1\perp}, p_{2\parallel})$ and $(p_{1\perp}, p_{2\perp})$

6 3 3 3 ← Factor.

→ If one of these dipoles is along $\vec{R}$ or ⊥ to $\vec{R}$ then we will have to evaluate only two terms.

## Torque On A Dipole About Its Axis:

Recall that potential energy of a dipole is given as $U = -\vec{p}\cdot\vec{E} = -pE\cos\theta$, Now suppose we slowly rotate the dipole about its axis by applying an external torque $\tau_{ext}$ then:




$$\tau_{ext}\, d\theta = dU \quad \text{(CHIME)}$$

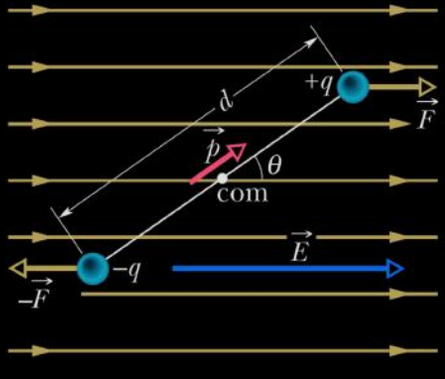
$$\Rightarrow \tau_{ext} = \frac{dU}{d\theta} = pE\sin\theta.$$

Since the dipole is rotated slowly, this external torque must be equal to the electrostatic torque thus electrostatic torque is also given by $\boxed{\tau = pE\sin\theta}$. This torque is restoring in nature, recall that least PE of dipole occurs for $\theta = 0°$. Vectorially the electrostatic torque can be written as:

$$\boxed{\vec{\tau} = \vec{p}\times\vec{E}}$$, This can be readily seen from the figure below, the electrostatic torque will try to align the dipole with the electric field, ∴ it must be clockwise for the shown figure, and $\vec{p}\times\vec{E}$ also has a clockwise sense.

→ For uniform fields even for elongated dipoles we can write torque as $\vec{p}\times\vec{E}$. This can be readily seen from adjacent figure.

$\tau = qE\frac{d}{2}\sin\theta \times 2$ $(\because \tau_\perp \times F)$.




YOUTUBE.COM/@THEPHYSICSSHELTER

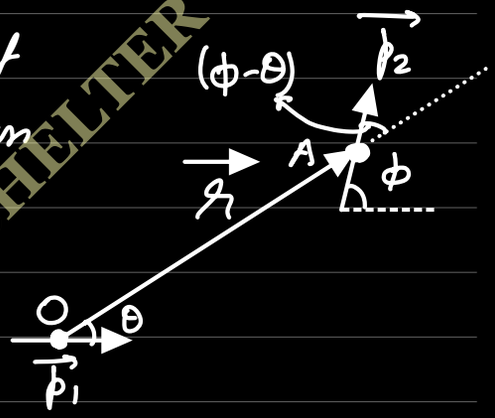
$\Rightarrow \tau = pE\sin\theta$ or $\vec{\tau} = \vec{p} \times \vec{E}$.

→ In non uniform fields, the forces on dipole charges do not form a couple, thus torque will depend on the point about which torque is being taken. There in addition to $\vec{p} \times \vec{E}$ one has to consider the torque due to $(\vec{p} \cdot \vec{\nabla})\vec{E}$ thus net torque will be given as:

$$\boxed{\vec{\tau} = \vec{p} \times \vec{E} + \vec{r} \times (\vec{p} \cdot \vec{\nabla})\vec{E}}$$

## Interaction Potential Energy Of Dipoles:

Let $\vec{p}_1$ and $\vec{p}_2$ be two point dipoles such that $\vec{p}_2$ is located at $(r, \theta)$ wrt $p_1$ and makes an angle $\phi$ with $p_1$. Now field due to $\vec{p}_1$ at point A:




$$\vec{E} = \frac{2kp_1\cos\theta}{r^3}\hat{r} + \frac{kp_1\sin\theta}{r^3}\hat{\theta}$$

also $\vec{p}_2 = p_2\cos(\phi-\theta)\hat{r} + p_2\sin(\phi-\theta)\hat{\theta}$

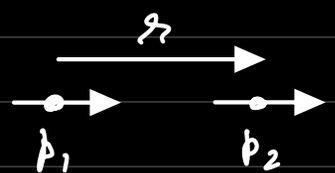
$$\Rightarrow U = -\vec{p}_2 \cdot \vec{E} = -\frac{kp_1p_2}{r^3}\left[2\cos\theta\cos(\phi-\theta) + \sin\theta\sin(\phi-\theta)\right] \quad (1)$$

The above expression can be used for expressing potential energies for various configuration of the dipoles.

## PE of Coaxial Dipoles:

Here $\theta = 0°$, $\phi = 0°$

$$\Rightarrow \boxed{U = \frac{-2kp_1p_2}{r^3}}$$

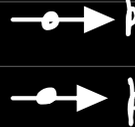



→ In case of antiparallel dipoles, $U = +\frac{2kp_1p_2}{r^3}$

→ This could also be derived directly using $-\vec{p} \cdot \vec{E}$ in coaxial position.

## PE of Parallel Dipoles ($\vec{r} \perp \vec{p}$)

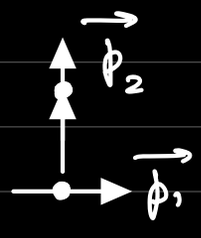
Here, $\theta = 90°$, $\phi = 0°$ $\Rightarrow \boxed{U = \frac{+kp_1p_2}{r^3}}$




YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

→ This could also be derived directly using $U = -\vec{p}\cdot\vec{E}$ for equatorial field.

## PE of ⊥ Dipoles ($\vec{r} \perp \vec{p}$, T shape config):




Here, $\theta = 90°$, $\phi = 90°$

$\Rightarrow \boxed{U = 0}$

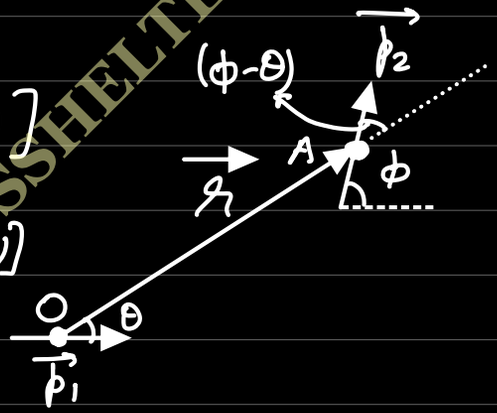
→ This could also be directly derived using $U = -\vec{p}\cdot\vec{E}$ in the equatorial configuration.

## Force Between Arbitrary Coplanar Dipoles:

From equation (1) above:

$$U = \frac{-kp_1p_2}{r^3}\left[2\cos\theta\cos(\phi-\theta) + \sin\theta\sin(\phi-\theta)\right]$$

$$\Rightarrow F_r = -\frac{\partial U}{\partial r} = \frac{3kp_1p_2}{r^4}\left[2\cos\theta\cos(\phi-\theta) + \sin\theta\sin(\phi-\theta)\right]$$




$$F_\theta = \frac{-1}{r}\frac{\partial U}{\partial \theta} = \frac{kp_1p_2}{r^4}\Big[2\cos\theta\times(-\sin(\phi-\theta))\times(-1) + 2\cos(\phi-\theta)[-\sin\theta] + \sin\theta\times\cos(\phi-\theta)\times(-1) + \sin(\phi-\theta)\times\cos\theta\Big]$$

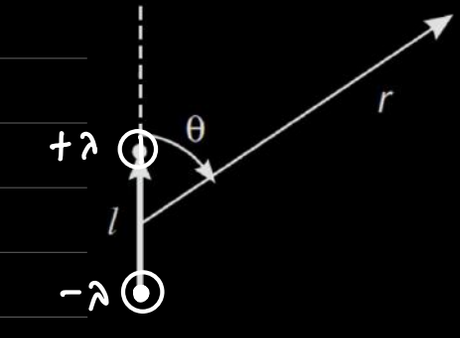
$$\Rightarrow F_\theta = \frac{kp_1p_2}{r^4}\left[3\cos\theta\sin(\phi-\theta) - 3\sin\theta\cos(\phi-\theta)\right]$$

$$\Rightarrow F_\theta = \frac{3kp_1p_2}{r^4}\sin(\phi-2\theta)$$

## Example: Irodov 3.42

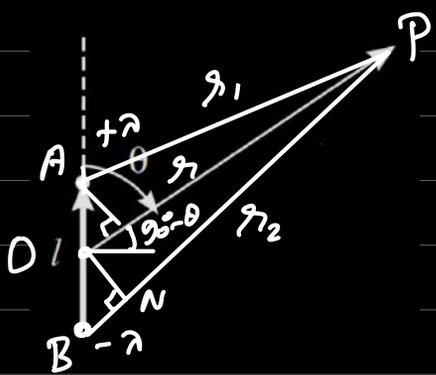
Two thin parallel threads carry a uniform charge with linear densities $\lambda$ and $-\lambda$. The distance between the threads is equal to $l$. Find the potential of the electric field and the magnitude of its strength vector at the distance $r \gg l$ at the angle $\theta$ to the vector $l$ (Fig. 3.5).

Solution: Here we shall try to find the field by differentiating the potential function. In the figure a projection of situation is shown where wires appear like points.




YOUTUBE.COM/@THEPHYSICSSHELTER

Now we place the compass at P and make an arc with radius PO to intersect BP at N.




$\Rightarrow BP = r_2 = r + \frac{l}{2}\cos\theta$, similarly

$$AP = r - \frac{l}{2}\cos\theta,$$

Recall potential due to an infinite wire of charge is given by $V = k\lambda \ln(r_0/r)$, thus by superposition, net potential at P:

$$V = 2k\lambda \ln\left(\frac{r_0}{r - \frac{l}{2}\cos\theta}\right) - 2k\lambda \ln\left(\frac{r_0}{r + l/2\cos\theta}\right)$$

$\Rightarrow V = 2k\lambda \ln\left(\frac{r + l/2\cos\theta}{r - l/2\cos\theta}\right)$, We would like to apply binomial since here

$\therefore$ We write $V = 2k\lambda \ln\left[\left(1 + \frac{l}{2r}\cos\theta\right)\left(1 - \frac{l}{2r}\cos\theta\right)^{-1}\right]$

$\Rightarrow V \approx 2k\lambda \ln\left[1 + \frac{l}{r}\cos\theta\right]$ ($\because$ Neglected Second order terms).

Now $E_r = \frac{-\partial V}{\partial r} = -2k\lambda \times \frac{1}{\left(1 + \frac{l}{r}\cos\theta\right)} \times \left(\frac{-l\cos\theta}{r^2}\right)$

$\Rightarrow E_r \approx \frac{2k\lambda l\cos\theta}{r^2}$

Now $E_\theta = \frac{-1}{r}\frac{\partial V}{\partial \theta} = \frac{2k\lambda}{r\left(1 + \frac{l}{r}\cos\theta\right)} \cdot \left(\frac{-l}{r}\right)\sin\theta \approx \frac{-2k\lambda l\sin\theta}{r^2}$

Net $E = \sqrt{E_r^2 + E_\theta^2} = \frac{2k\lambda l}{r^2}$ Ans

Note: $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} \ldots$ thus we could Also write

$$V = 2k\lambda \ln\left(1 + \frac{l}{r}\cos\theta\right) \approx 2\frac{k\lambda l}{r}\cos\theta \text{ Ans}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

## EXAMPLE: PATHFINDER MCQ 24

24. Three identical electric dipoles are arranged parallel to each other at equal separations as shown in the figure. The separation between the charges of a dipole is negligible as compared to the separation between the dipoles. In the given configuration, total electrostatic interaction energy of these dipoles is $U_0$. Now one of the end dipole is gradually reversed, how much work is done by the electric forces?

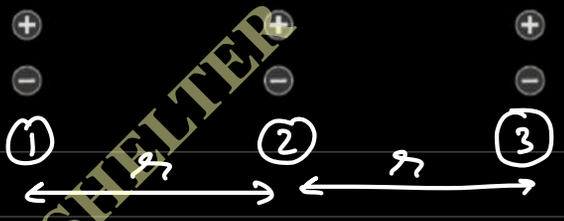


(a) $\frac{17U_0}{18}$ (b) $-\frac{17U_0}{18}$

(c) $\frac{18U_0}{17}$ (d) $-\frac{18U_0}{17}$

Recall interaction energy for dipoles in parallel ($p \perp r$) configuration is:

$$U = \frac{kp_1p_2}{r^3}$$




Initially interaction energy: $U_{12} + U_{23} + U_{31}$

$$\Rightarrow U_i = \frac{kp^2}{r^3}\left[\frac{1}{1} + \frac{1}{1} + \frac{1}{2^3}\right] = \frac{17}{8}\frac{kp^2}{r^3} = U_0$$

Now when the end dipole is reversed, its interaction energy with other dipoles changes sign.

$$\Rightarrow U_f = \frac{kp^2}{r^3}\left[1 - 1 - \frac{1}{2^3}\right] = -\frac{kp^2}{8r^3}$$

$$\Rightarrow \Delta W_E = -\Delta U = -\left(-\frac{1}{8} - \frac{17}{8}\right)\frac{kp^2}{r^3} = \frac{+18}{8}\frac{kp^2}{r^3} = \frac{+18}{8} \times \frac{8}{17} U_0 = \frac{+18}{17} U_0$$

$\therefore$ Option (c)

**QUADRUPOLES AND OCTOPOLES:** We can think of dipole as being obtained by including a negative monopole (charge)




| Monopole | Dipole | Quadrupole | Octopole |
|---|---|---|---|
| $(V \sim 1/r)$ | $(V \sim 1/r^2)$ | $(V \sim 1/r^3)$ | $(V \sim 1/r^4)$ |

YOUTUBE.COM/@THEPHYSICSSHELTER

Similarly if we include a negative of a dipole (opposite dipole moment), we get a Quadrupole. Similarly if we include another Quadrupole (see last figure, total 8 point charges) we get an Octopole.

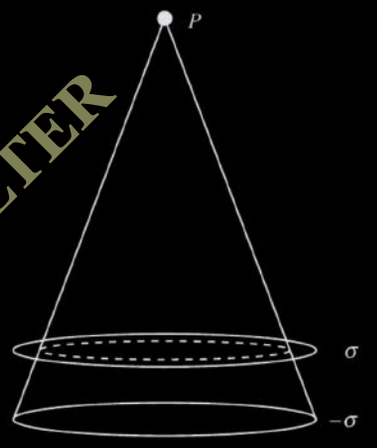
Just as we showed for a point dipole that $V \propto 1/r^2$ and $E \propto 1/r^3$ similarly we can show that for a Quadrupole, $V \propto 1/r^3$ and $E \propto 1/r^4$ and for an Octopole, $V \propto 1/r^4$ and $E \propto 1/r^5$.

## EXAMPLE PURCELL AND MORIN 2.16

Two parallel disks each have radius $R$ and are separated by a distance $\ell$. The surface charge densities are $\sigma$ and $-\sigma$. What is the electric field at a large distance $r$ along the axis of the disks? Solve this in two ways.

(a) Treat the disks like a collection of a large number of dipoles standing next to each other.

(b) Explain why the parts of the two disks that are contained within the cone in Fig. 2.45 produce canceling fields at point $P$, and then find the field due to the uncanceled part of the top disk.




Solution: Method 1: Dipole moment (using separation of centres of charges)

$\vec{p} = \sigma \times \pi R^2 \ell \hat{j}$, Now field in axial position:

$$E = \frac{2kp}{r^3} = 2\pi \frac{1}{4\pi\epsilon_0} \times \frac{\sigma \times \pi R^2 \ell}{r^3}$$

$$\Rightarrow \quad E = \frac{\sigma}{2\epsilon_0} \frac{R^2 \ell}{r^3} \quad \text{Ans.}$$

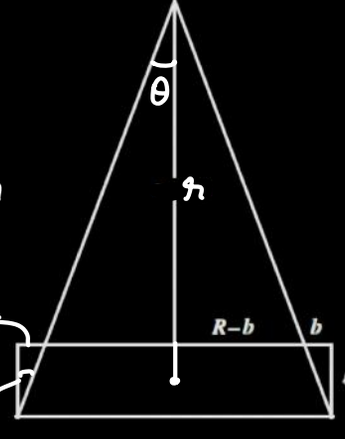
Method 2: Recall that normal field created by a lamina is $k\sigma\omega$ now $\omega$ is same for lower disc and the portion of the upper disc within dotted portion. Thus these two will give cancelling fields and we only need to consider the field due to the annular region outside the dotted circle, which behaves like a ring.

Now field due to a circle of charge:

$$E = \frac{kQr}{(R^2+r^2)^{3/2}}, \text{ In our case,}$$

$$E = \frac{k \times \sigma \times 2\pi R \times \ell \tan\theta \times r}{r^3} \qquad \tan\theta \approx R/r$$

$$= \frac{\sigma}{2\epsilon_0} \frac{R^2 \ell}{r^3} \quad \text{Ans.}$$




YOUTUBE.COM/@THEPHYSICSSHELTER

**EXAMPLE:** A dipole is to be placed with its axis along axis of a uniformly charged ring. Where will the stable equilibrium position occur? Suppose a dipole of mass $m$ and dipole moment $p$ is kept at such a position for a ring with radius $R\sqrt{2}$ and charge $Q$. Will such a dipole execute SHM. If not then to what power of amplitude will the time period be proportional to?

Solution: Recall that potential energy of dipole is $-\vec{p}\cdot\vec{E}$, thus stable equilibrium occurs when $\vec{p} \parallel \vec{E}$ and $E$ is maximum. (This is configuration of lowest PE). ∴ If ring has radius $R$, then dipole can oscillate about point $x = R/\sqrt{2}$.

(b) $E = \dfrac{kQx}{(R^2+x^2)^{3/2}}$, for our ring $R \to \sqrt{2}R$ and at stable position, $x = \sqrt{2}R/\sqrt{2} = R$, Now we need to find PE upon small displacement $y$ from this position to evaluate effective spring constant for the situation. PE at small distance $y$ from equilibrium position:

$$U = \frac{-kQ(R+y)p}{[2R^2+(R+y)^2]^{3/2}}, \quad y \ll R, \text{ Now we shall}$$

do a binomial approximation to find out how the potential Energy varies with $y$. For SHM to occur, the change in potential energy should be proportional to $y^2$ (∵ Just like spring energy change is $\frac{1}{2}kx^2$). ∵ We only need to watch out the powers of $y$ for figuring out nature of oscillations we might as well put $R$, $Q$ and $p$ each to be one unit. Thus we need to apply binomial theorem to expression:

$$\frac{-(1+y)}{[2+(1+y)^2]^{3/2}} \simeq \frac{-(1+y)}{(2+1+2y)^{3/2}} = \frac{-(1+y)}{3^{3/2}\left(1+\frac{2}{3}y\right)^{3/2}}$$

$$= \frac{-(1+y)(1-y)}{3^{3/2}} = \frac{-1}{3^{3/2}}(1-y^2) = \frac{1}{3^{3/2}}(-1+y^2)$$

Thus potential energy is of the form $U_0 + Cy^2$, This is of the same form as spring potential energy (i.e $U_0 + \frac{1}{2}kx^2$), thus motion must be an SHM.

YOUTUBE.COM/@THEPHYSICSSHELTER

## Binomial Expansion For Negative And Fractional Powers:

An expression of the form $(1+x)^n$ can be expanded as:

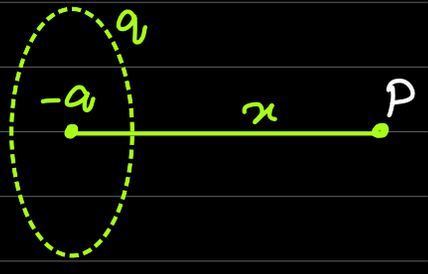
$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots \text{ infinite terms.}$$

Most of the time we just need first order term but if upon simplification of an expression, first order term vanishes, we might need expansion up to second order term.

## Example: 3.10 (IRODOV)

A point charge $q$ is located at the centre of a thin ring of radius $R$ with uniformly distributed charge $-q$. Find the magnitude of the electric field strength vector at the point lying on the axis of the ring at a distance $x$ from its centre, if $x \gg R$.

Solution: Here net dipole moment is zero so we can't use the result of a standard dipole.




Now:

$$E = \frac{kqx}{(R^2+x^2)^{3/2}} - \frac{kq}{x^2}$$

$$\Rightarrow E = \frac{kq}{x^2}\left[\frac{1}{[1+(R/x)^2]^{3/2}} - 1\right] = \frac{kq}{x^2}\left[(1+(R/x)^2)^{-3/2} - 1\right]$$

$$\Rightarrow E = \frac{kq}{x^2}\left[1 - \frac{3}{2}\frac{R^2}{x^2} - 1\right] = \frac{-3kqR^2}{2x^4} = \frac{-3qR^2}{8\pi\epsilon_0 x^4}$$

$$\Rightarrow |E| = \frac{3qR^2}{8\pi\epsilon_0 x^4}$$

Alternate Solution: $V(x) = \frac{-kq}{x} + \frac{kq}{\sqrt{R^2+x^2}} = \frac{kq}{x}\left[\frac{1}{\sqrt{1+(R/x)^2}} - 1\right]$

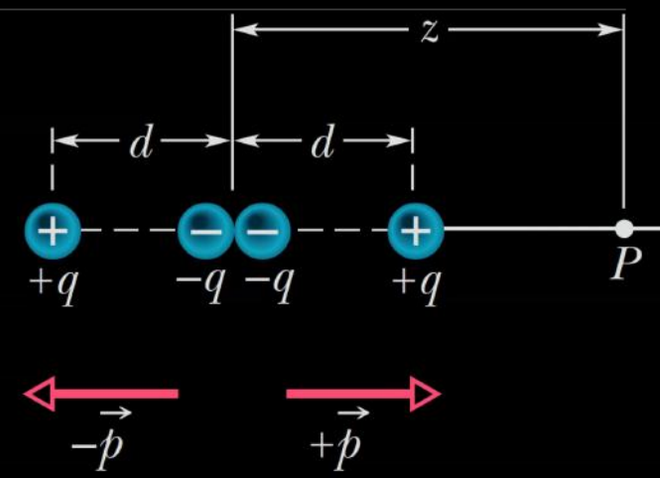
$$\Rightarrow V(x) = \frac{kq}{x}\left[(1+(R/x)^2)^{-1/2} - 1\right]$$

$$= \frac{kq}{x}\left[1 - \frac{1}{2}\left(\frac{R}{x}\right)^2 - 1\right]$$

$$= \frac{-kqR^2}{2x^3}, \text{ Now } E = -\frac{\partial V}{\partial x} = \frac{kqR^2}{2}\left[-3x^{-4}\right] = \frac{-3qR^2}{8\pi\epsilon_0 x^4} \text{ Ans.}$$

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

## EXAMPLE (WALKER): (ELECTRIC QUADRUPOLE)




The adjacent figure shows a generic electric Quadrupole. It consists of two dipoles with dipole moments equal in magnitude but opposite in directions as shown. Show that value of E on axis of the quadrupole for a point P at at distance z from its centre is given by

$$E = \frac{3Q}{4\pi\epsilon_0 z^4} \quad \text{where } Q = 2qd^2$$

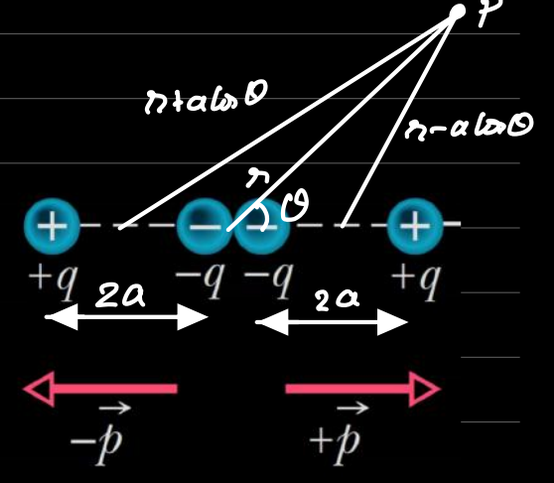
known as Quadrupole moment for the charge distribution.

Solution: By superposing the field due to two dipoles:

$$E = \frac{2kp}{(z-\frac{d}{2})^3} - \frac{2kp}{(z+\frac{d}{2})^3} = \frac{2kp}{z^3}\left[\left(1-\frac{d}{2z}\right)^{-3} - \left(1+\frac{d}{2z}\right)^{-3}\right]$$

$$\Rightarrow E = \frac{2kp}{z^3}\left[\frac{3d}{z}\right] = \frac{2}{3^4} \times \frac{1}{4\pi\epsilon_0} \times qd^2 \times 3 = \frac{3Q}{4\pi\epsilon_0 z^4} \text{ Ans.}$$

**EXAMPLE:** For the quadrupole of previous problem, find potential at a general $(r, \theta)$.




Solution: Recall that potential due to a dipole is given as:

$$V = \frac{kp\cos\theta}{r^2}, \text{ Using this}$$

for the shown two dipoles,

$$V = \frac{kp\cos\theta}{(r-a\cos\theta)^2} - \frac{kp\cos\theta}{(r+a\cos\theta)^2}$$

$$\Rightarrow V = \frac{kp}{r^2}\left[\left(1-\frac{a\cos\theta}{r}\right)^{-2} - \left(1+\frac{a\cos\theta}{r}\right)^{-2}\right] = \frac{4kpa\cos\theta}{r^3} \text{ Ans.}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

→ Now if we want we can find the field using $E_r = \frac{-\partial V}{\partial r}$

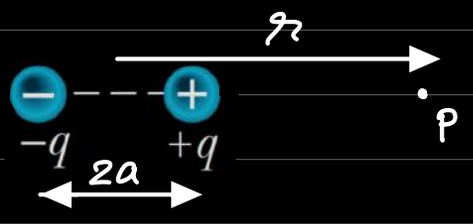
and $E_\theta = \frac{-1}{r}\frac{\partial V}{\partial \theta}$.

→ Field of a Quadrupole varies as $1/r^4$.

→ In a similar manner it can be shown that potential of an octupole varies as $1/r^4$ and field as $1/r^5$ (Similarly the pattern can be continued for a Hexadecapole etc).

**EXAMPLE:** Recall the axial field due to a dipole when written as $E = 2kp/r^3$ involves an approximation. A better estimate can be obtained by considering still higher order terms. Try to come up with a slightly better formula by estimating next higher order term.

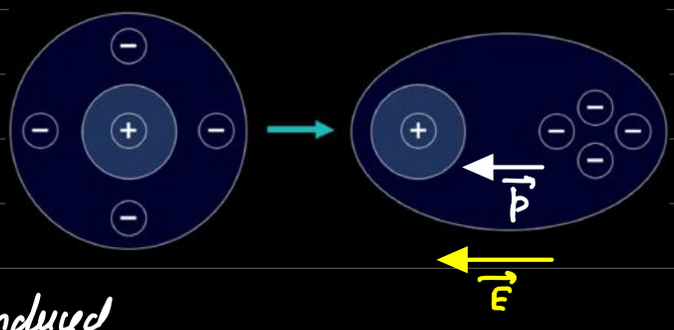
Solution: $E = \frac{kq}{(r-a)^2} - \frac{kq}{(r+a)^2}$




$$\Rightarrow E = \frac{kq}{r^2}\left[\frac{4a/r}{\left(1-\frac{a^2}{r^2}\right)}\right]$$

$$= \frac{4kqa}{r^3}\left[1+\frac{a^2}{r^2}\right] = \frac{4kqa}{r^3} + \frac{4kqa^3}{r^5} = \frac{2kp}{r^3} + \frac{2kpa^2}{r^5}$$

**POLARIZABILITY:** The tendency of a non polar molecule to form a dipole moment when exposed to an external electric field is called its polarizability ($\beta$). Mathematically it is quantified as dipole moment induced per unit electric field or $\boxed{\vec{p} = \beta\vec{E}}$




YOUTUBE.COM/@THEPHYSICSSHELTER

**EXAMPLE: IRODOV 3.73**

A non-polar molecule is located at the axis of a thin uniformly charged ring of radius $R$. At what distance $x$ from the ring's centre is the magnitude of the force $F$ acting on the given molecule
(a) equal to zero; (b) maximum?
Draw the approximate plot $F_x(x)$. (Assume constant polarizability).

**Solution:** For dipole along axis of ring, $U = -\vec{p}\cdot\vec{E} = -pE$.

$\Rightarrow U = -\beta E^2 \Rightarrow F = -\frac{\partial U}{\partial x} = 2\beta E\frac{dE}{dx}$, this force will attain a minimum value where $\frac{dE}{dx} = 0$ i.e $x = R/\sqrt{2}$. Thus force is minimum at $x = R\sqrt{2}$.

Now: $E = \frac{kQx}{(R^2+x^2)^{3/2}}$, let $x = \eta R$ where $\eta$ is a dimensionless number $\Rightarrow E^2 \propto \frac{(\eta R)^2}{(R^2+(\eta R)^2)^3}$ i.e $\frac{\eta^2}{(1+\eta^2)^3}$ ①

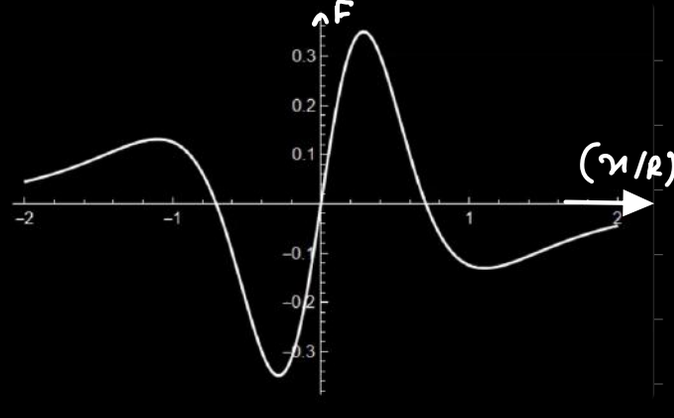
Now For maximum force, $\frac{d^2(E^2)}{d\eta^2} = 0$

(Actually we wanted $\frac{d^2(E^2)}{dx^2} = 0 \Rightarrow \frac{d}{d\eta}\left[\frac{dE^2}{d\eta}\times\frac{d\eta}{dx}\right]\left(\frac{d\eta}{dx}\right) = \frac{1}{R^2}\frac{d^2(E^2)}{d^2\eta}$

Now Differentiating ① twice and equating to zero, we get:

$$\frac{20\eta^4 - 26\eta^2 + 2}{(1+\eta^2)^5} = 0 \Rightarrow \eta = \pm\frac{1}{2}\sqrt{\frac{13\pm\sqrt{129}}{5}} \text{ dm.}$$

A qualitative plot of relative magnitude of F with the ratio $\eta$ i.e $x/R$ looks like shown in the adjacent figure.




Thus maximum magnitude of force occurs at the lower magnitude root i.e $\pm\frac{1}{2}\sqrt{\frac{13-\sqrt{129}}{5}}$

YOUTUBE.COM/@THEPHYSICSSHELTER

**EXAMPLE: IRODOV 3.72 :**

A non-polar molecule with polarizability $\beta$ is located at a great distance $l$ from a polar molecule with electric moment $\mathbf{p}$. Find the magnitude of the interaction force between the molecules if the vector $\mathbf{p}$ is oriented along a straight line passing through both molecules.

Solution: Dipole moment of the non polar molecule, $\vec{p'} = \beta \vec{E}$

$\Rightarrow U = -\vec{p}\cdot\vec{E} = -\beta E^2 = -\beta\left(\frac{2kp}{r^3}\right)^2 = -\frac{4\beta k^2 p^2}{r^6}$

$\Rightarrow F = -\frac{\partial U}{\partial r} = \frac{-24\beta k^2 p^2}{r^7}$ Ans. (Note that force between two polar molecules was proportional to $1/r^4$, whereas here it is proportional to $1/r^7$).

YOUTUBE.COM/@THEPHYSICSSHELTER